

MESSOU FRANCK

AI / Machine Learning Engineer • LLMs & GenAI • NLP & Speech • MLOps & Model Serving

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📍 Tokyo, Japan

in [mesabo](#)

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🔗 [mesabo.github.io](#)

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Machine learning engineer and PhD researcher with 3+ years of experience shipping applied AI end-to-end: on-device fine-tuned small language models with cloud LLM-API fallback (Gemini, GPT), speech and NLP pipelines (ASR, summarization, translation across 20 languages), and IoT / edge sensor intelligence. Recently built **AIReco**, an AI meeting-assistant, and shipped **CVFlow** ([cvflow.app](#)), a 20-language AI resume platform. First-author of **2 Q1 journals and 5 IEEE conference papers**, with a solid MLOps foundation: LoRA / PEFT fine-tuning, distributed training (DDP), and low-latency vLLM serving on multi-GPU infrastructure, plus LangChain / LangGraph / RAG pipelines. Available immediately for a **machine learning or applied-AI engineering role** to take models from research to production.

KEY ACHIEVEMENTS

- Built **AIReco**, an AI meeting-assistant: an offline sLM fine-tuned with QLoRA plus cloud LLM APIs for transcription, summarization, and translation across 20 languages.
- Shipped **CVFlow** ([cvflow.app](#)), integrating Gemini / GPT for ATS scoring and AI generation across a **20-language UI**.
- First-author of **2 Q1 journals and 5 IEEE conference papers** (GLOBECOM, ICC, VTC); IEEE VTS 2024 Student Paper Award.
- Provisioned and operate a multi-GPU server end-to-end: distributed training (DDP), checkpointing, and vLLM serving in Docker.

EXPERIENCE

LLM Systems Engineer

Open-Source / Personal Research Initiative

📅 2025 – Present

📍 Remote • [GitHub.com](#)

- Architected and deployed four open-source production LLM systems (semantic search, retrieval-augmented generation, stateful agent orchestration, full-stack LLMOps), shipped via **GitHub Actions CI/CD** to GCP (Cloud Run, GKE, Vertex AI).
- Engineered composable LCEL pipelines with hybrid retrieval (FAISS, ChromaDB, BM25 sparse, MMR re-ranking), evaluated with RAGAS (context recall, faithfulness, semantic similarity).
- Built stateful LangGraph agents (conditional routing, cycles, human-in-the-loop) and ReAct loops with Pydantic-validated tools; FastAPI streaming APIs with semantic caching and LangSmith tracing at sub-200ms TTFT.

LangChain

LangGraph

LangSmith

FAISS

ChromaDB

RAGAS

GCP

Machine Learning Engineer (AIoT)

D'isum Inc.

📅 Oct 2025 – Jul 2026

📍 Tokyo, Japan — Hybrid

- Built ETL and preprocessing pipelines converting raw sensor streams from **50+ engines and 200+ batteries** into analytics-ready datasets (cleaning, segmentation, calibrated chunking).
- Developed and deployed classical ML models for engine sound analysis and battery time-series health monitoring (anomaly and fault detection).
- Designed feature-engineering and evaluation pipelines, benchmarking model families to select the best detector per signal type.

PROJECTS

SmolSearch – Semantic Search API

FAISS • LCEL • FastAPI • Cloud Run • [Github.com](#)

- ★ LCEL chain (SmolLM2-360M) over a FAISS index; FastAPI on Cloud Run, sub-100ms TTFT, MMR search.

RAGify – Evaluated RAG System

ChromaDB • RAGAS • GCP • [Github.com](#)

- ★ Hybrid retrieval (ChromaDB, FAISS, MultiQuery, BM25) scored with RAGAS on GCP.

AgentFlow – Stateful Multi-Tool Agent

LangGraph • ReAct • LangSmith • GKE • [Github.com](#)

- ★ LangGraph StateGraph with Pydantic tools and full LangSmith tracing; deployed on GKE.

LLMOps Baseline – Production LLMOps

Reference

LangChain • LangSmith • Vertex AI • FastAPI •

[Github.com](#)

- ★ LCEL + Pydantic output, Redis caching, SSE streaming, tiktoken costs; **GitHub Actions CI/CD** to Vertex AI.

SensoryHub – IoT Predictive Maintenance

SaaS

Team project • 2026 • Remote

- Architected a cloud-native IoT platform end-to-end: ESP32 sensor-node firmware and Raspberry Pi gateways streaming via MQTT into Python / FastAPI microservices (TimescaleDB, Redis Streams).
- ML anomaly detection (Isolation Forest, XGBoost) served via ONNX, with LLM root-cause analysis (SmolLM2) and RAG on the roadmap.
- React dashboards (admin + management), a sensor-capture app, multi-tenant auth (Keycloak), and Kubernetes / Terraform infrastructure.

TECHNICAL SKILLS

Python

PyTorch

Hugging Face

LLM Fine-tuning

LoRA / PEFT

LangChain

LangGraph

LangSmith

Text in blue is clickable – it opens the linked profile, project, or paper.

- Built on-device iOS / Android sensor apps (Swift, Java) for real-time data capture and edge inference.
- Deployed models to client systems via Azure cloud services, with input validation, error handling, and monitoring.

Python scikit-learn Swift Java Azure

Generative AI Intern (Summer Internship)

日本AIコンサルティング (Japan AI Consulting)

📅 Summer 2024 📍 Osaka, Japan

- Developed and maintained log data processing pipelines using FastAPI and Flask, with technical documentation managed via Growi.
- Customized AI model source code and integrated LLMs into interactive UIs using Gradio, delivering tailored outputs for internal use cases.
- Produced research reports on data privacy and the AI industry landscape, contributing insights to support J-AIC's business expansion strategy.

FastAPI Flask Gradio PostgreSQL LLMs

AI / ML Researcher (Doctoral)

YuLab, Hosei University

📅 Sep 2025 – Present 📍 Tokyo, Japan

- Built LLM / sLM reprogramming pipelines (Spec2Llama, Spec2LLM, LLM4Imp) adapting frozen LLMs and compact backbones (SmolLM, Phi, Qwen) for forecasting and imputation on edge, updating **under 9% of parameters** while matching or beating SAITS / TimeMixer baselines; published at IEEE ICCE / ICC 2026.
- Designed graph-constrained multi-LLM ensembles with structural causal models, lifting causal-discovery **F1 to 0.945** (+11 pts over PCMCi) and narrative completeness **5.6×** over a single-LLM baseline.

PyTorch Hugging Face LoRA / PEFT vLLM CUDA Docker

ML Researcher (Master's)

YuLab, Hosei University

📅 2023 – Sep 2025 📍 Tokyo, Japan

- Built transformer time-series forecasting models (TAPformer, TSformer, CNN-BiLSTM) for long-term electricity load forecasting, outperforming standard CNN / LSTM baselines (published in *Internet of Things*, Q1).

PyTorch Transformers Time-Series GNN

Applied AI Engineer – AIReco

Independent

📅 2025 – 2026 📍 Remote

- Built the ML core of a Flutter meeting-assistant: hybrid offline QLoRA-tuned sLM plus cloud LLM APIs for speech-to-text, summarization, and translation across 20 languages on a GCP backend.

Flutter GCP QLoRA ASR

RAG RAGAS FAISS vLLM Gemini
GPT ASR NMT Time-Series
Anomaly Detection GNN FastAPI
Docker DDP GCP Vertex AI

SOFT SKILLS

End-to-End Ownership Mentoring
Remote Collaboration
Cross-cultural Teamwork
Research Mindset Communication

OWNERSHIP

Co-Founder & Full-Stack Lead

CVFlow (cvflow.app)

📅 2025 – Present 📍 Remote

- Co-built a monetized AI resume platform; owned the AI integration (Gemini / GPT) and product.

Engineering Lead

EKKOU Platform (commercial)

📅 2024 – Present 📍 Tokyo

- Lead a full-stack AI platform: clients, APIs, GPU model serving, and CI/CD.

EDUCATION

PhD, Applied Informatics

Hosei University

📅 2025 – 2027 (Expected) 📍 Tokyo

MSc, Applied Informatics

Hosei University

📅 2023 – 2025 📍 Tokyo

AWARDS & CERTIFICATES

- IEEE VTS Tokyo / Japan Chapter 2024 Student Paper Award & Encouragement Award (VTC2024-Fall).
- JICA Development Studies Program (JICA専用講座), 2025.
- Reviewer: IEEE ICECET 2026 (Italy Section); Theoretical Computer Science (Elsevier).

LANGUAGES

English ●●●●●
French ●●●●●
Japanese ●●●●●